Unit 7, Lesson 1: Finding Slope Using Graphs and Tables

Benchmark

MA.8.AR.3.2 Given a table, graph, or written description of a linear relationship, determine the slope.

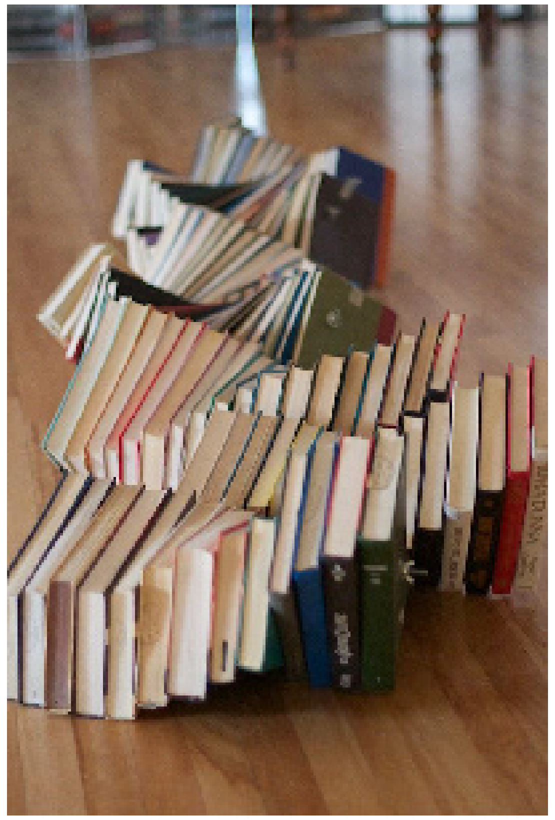
Learning Targets

- ☐ I can determine the slope of a line from a graph or a table, given a mathematical context.
- ☐ I can relate the slope as the rate of change of the relationship between variables.

7.1.1 Warm-Up: Would You Rather?

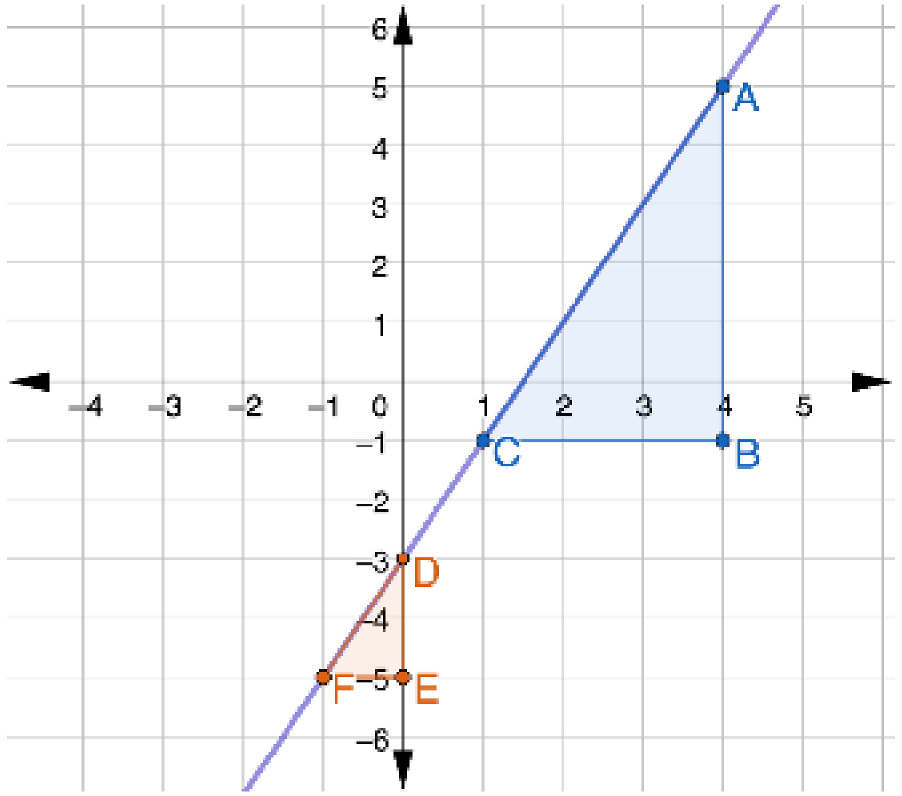
1. Would you rather read 12 pages each night in a book with 144 total pages or read 50 pages three times a week in a book with 132 total pages?

   Explain your reasoning using mathematics for whichever option you choose.



7.1.2 Refresher: Using Similar Triangles to Find the Slope of a Line

Similar triangles can be used to determine the slope of a line by comparing the ratio of vertical lengths to horizontal lengths of the triangles.



1. For each triangle, label the ordered pairs of each vertex.

2. What is the length of the vertical leg of the blue triangle?

3. How could you use the coordinates of the vertices of the vertical leg to find its length?

4. What is the length of the horizontal leg of the blue triangle?

5. How could you use the coordinates of the vertices of the horizontal leg to find its length?

The ***slope*** of a line is the ratio of the change in the vertical direction to change in the horizontal direction, often expressed as the $\frac{change\ in\ y}{change\ in\ x}$, or $\frac{\Delta y}{\Delta x}$.

6. Complete the statement.

In the equation $slope = \frac{change\ in\ y}{change\ in\ x}$, the [o change in y / o change in x] relates to finding the length of the vertical leg of the triangle and the [o change in y / o change in x] relates to finding the length of the horizontal leg of the triangle.

7. Use the coordinates of the vertices of the blue triangle to find the slope of the purple line. Show your work or explain what you did.

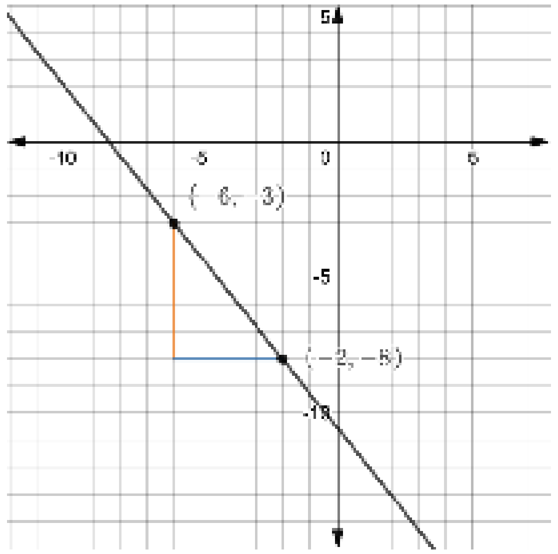
8. Use the coordinates of the vertices of the orange triangle to find the slope of the purple line. Show your work or explain what you did.

9. Check your answers with your partner.

10. Make a conjecture about whether the order in which you subtract to find the change in x and change in y matters.

7.1.3 Collaboration: Finding Slope from Graphs

Fatima and Oliver both drew the same triangle using the points $(-6,-3)$ and $(-2,-8)$ on the line to determine the slope, but they performed the calculations differently. Their partial work is shown.



1. With your partner, complete each student's work and answer the questions that follow.

Fatima's Work	Oliver's Work
$slope = \frac{\text{change in } y}{\text{change in } x}$ $slope = \frac{-3-(-8)}{-6-(-2)}$ $slope =$	$slope = \frac{\text{change in } y}{\text{change in } x}$ $slope = \frac{-8-(-3)}{-2-(-6)}$ $slope =$

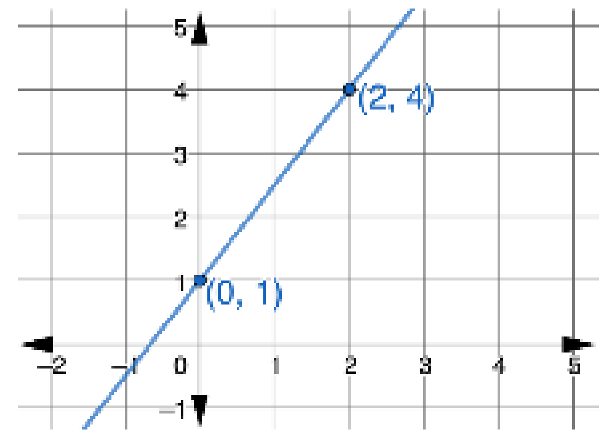
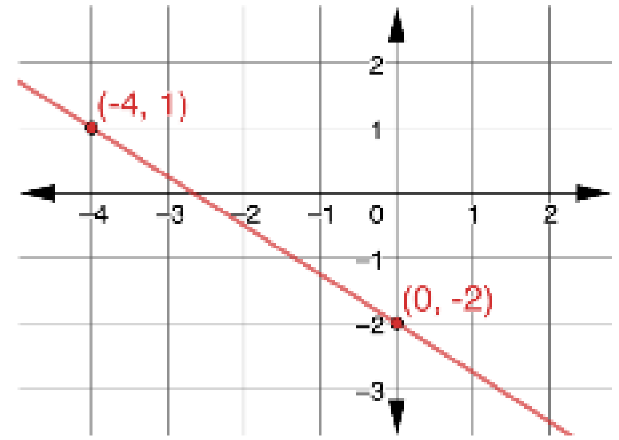
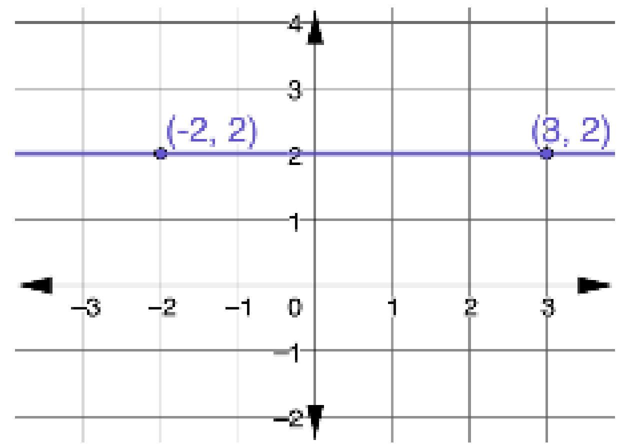
2. Fatima compared her work to Oliver's and said, "It doesn't matter which order we subtracted the values. We found the same slope." Explain why you agree or disagree.

3. Oliver said, "It doesn't matter which order you subtract, but whichever coordinate pair you start with to find the change in y, you have to also start with its x-value when finding the change in x." What do you think Oliver meant?

7.1.4 Guided Instruction: Finding Slope from Graphs and Tables

The slope of any line can be determined from a graph using any two points on the line.

Three graphs and the calculations used to determine their slope are shown.

Graph	Calculation	Slope
	$m = \frac{4-1}{2-0} = \frac{3}{2}$	$\frac{3}{2}$
	$m = \frac{1-(-2)}{-4-0} = \frac{3}{-4}$	$-\frac{3}{4}$
	$m = \frac{2-2}{3-(-2)} = \frac{0}{5}$	0

1. Use the graphs and their slopes to complete the summary.

The slope of a line can be positive, negative, or 0.

The slope is positive when the line
- o does not rise or fall
- o is increasing, or rising,
- o is decreasing, or falling,

from left to right.

The slope is negative when the line
- o does not rise or fall
- o is increasing, or rising,
- o is decreasing, or falling,

from left to right.

The slope is 0 when the line
- o does not rise or fall
- o is increasing, or rising,
- o is decreasing, or falling,

from left to right.

Recall that linear relationships can also be represented using tables of values.

The table to the right represents a linear relationship.

x	y
–4	–9
–2	–4
0	1
2	6
4	11

Notice that the x-values in the table are increasing by the same amount in each row.

2. How are the y-values in the table changing as the x-values increase by 2?

3. With your partner, decide who will be Partner A and who will be Partner B. Then, find the slope of the linear relationship using the ordered pairs indicated below.

   a. Show your work in the space provided.

Partner A	Partner B
Find the slope using the ordered pairs $(-2, -4)$ and $(0, 1)$ from the table.	Find the slope using the ordered pairs $(4, 11)$ and $(2, 6)$ from the table.

   b. Check your work with your partner.

The slope of any line can be determined from a table of values using any two ordered pairs in the linear relationship because the ***rate of change*** in a linear relationship is constant.

The ***rate of change*** is the ratio of change in one quantity to the corresponding change in another quantity.

A rate, such as 45 miles per hour, gives the speed of an object. The rate of change of a line tells us how fast or how slow a line is increasing or decreasing.

Rates can be constant. For example, the speed of a horse on a carousel is constant while the ride is in motion, but the speed of someone riding on a merry-go-round is not constant because it can change depending on how fast or slow the merry-go-round is being pushed.

4. Complete the summary to describe the relationships between the quantities observed for the linear relationship represented by the table of values.

The y-values in the table
- o increase by 2
- o increase by 5

each time the x-values
- o increase by 2.
- o increase by 5.

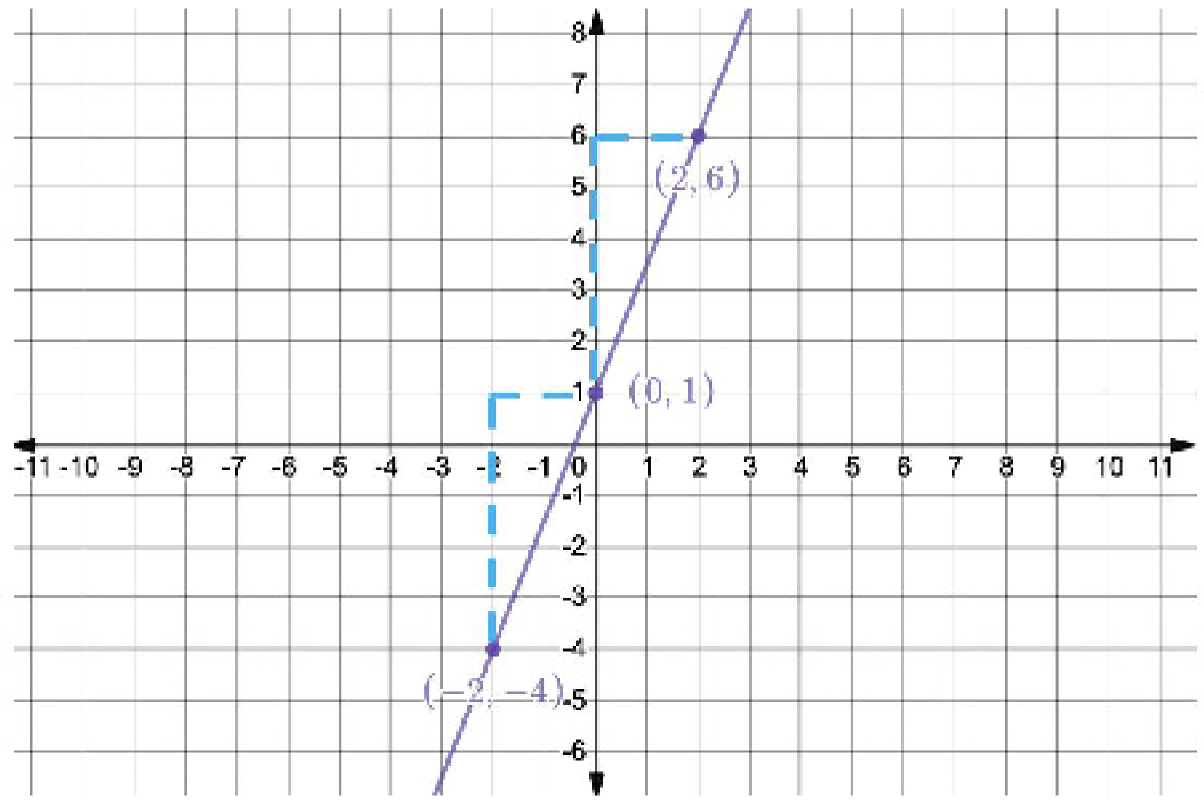
Slope is the
- o change in x
- o change in y

over the
- o change in x,
- o change in y,

or $\frac{\square}{\square}$ in this case. The slope, or rate of change, is constant for linear relationships, so the slope is
- o the same, regardless of
- o different, depending on

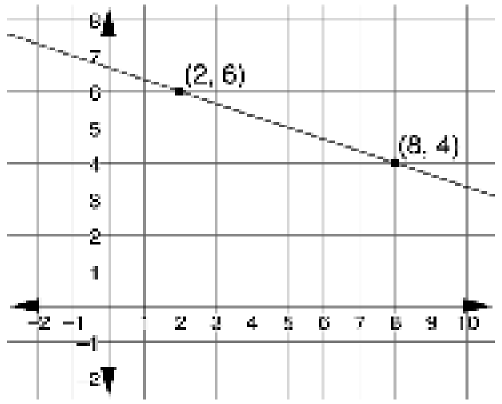
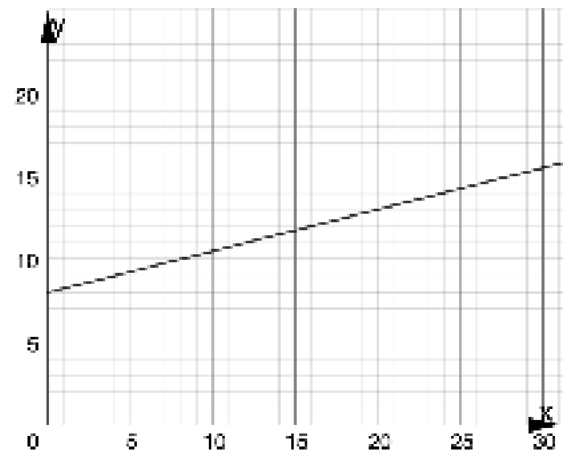
which two ordered pairs on the line are used to calculate it.

This means that the line is increasing or decreasing at the same rate between every point along the line. That is why it forms a straight line, as shown on the graph.



7.1.5 Your Turn!

1. Determine the slope of each linear relationship.

<table>
<tr><th>Linear Relationship</th><th>Calculation</th><th>Slope</th></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>−2</td><td>−11</td></tr><tr><td>−1</td><td>−8</td></tr><tr><td>0</td><td>−5</td></tr><tr><td>1</td><td>−2</td></tr></table></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td><table><tr><th>x</th><th>y</th></tr><tr><td>−3</td><td>−5</td></tr><tr><td>0</td><td>−5</td></tr><tr><td>3</td><td>−5</td></tr><tr><td>6</td><td>−5</td></tr></table></td><td></td><td></td></tr>
</table>

7.1.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can determine the slope of a line from a graph or a table, given a mathematical context.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can relate the slope to the rate of change of the relationship between variables.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

